

K

LV.0 → LV.1 · AI ENGINEER

12 K-words every Level 0 → 1 AI engineer must know.

KV Cache, Knowledge Distillation, K-Means & more. **Plain-English** definitions, one swipe each, plus the K-named labs and minds to follow.

↓ UP NEXT • TERMS

The K stack.

KV Cache

K-Means

KNN

Keras

Kafka

Kubernetes

Kurtosis

Knowledge Graph

Knowledge Base

K-Fold CV

Kernels

Knowledge Distillation

K

KURTOSIS • STATISTICS

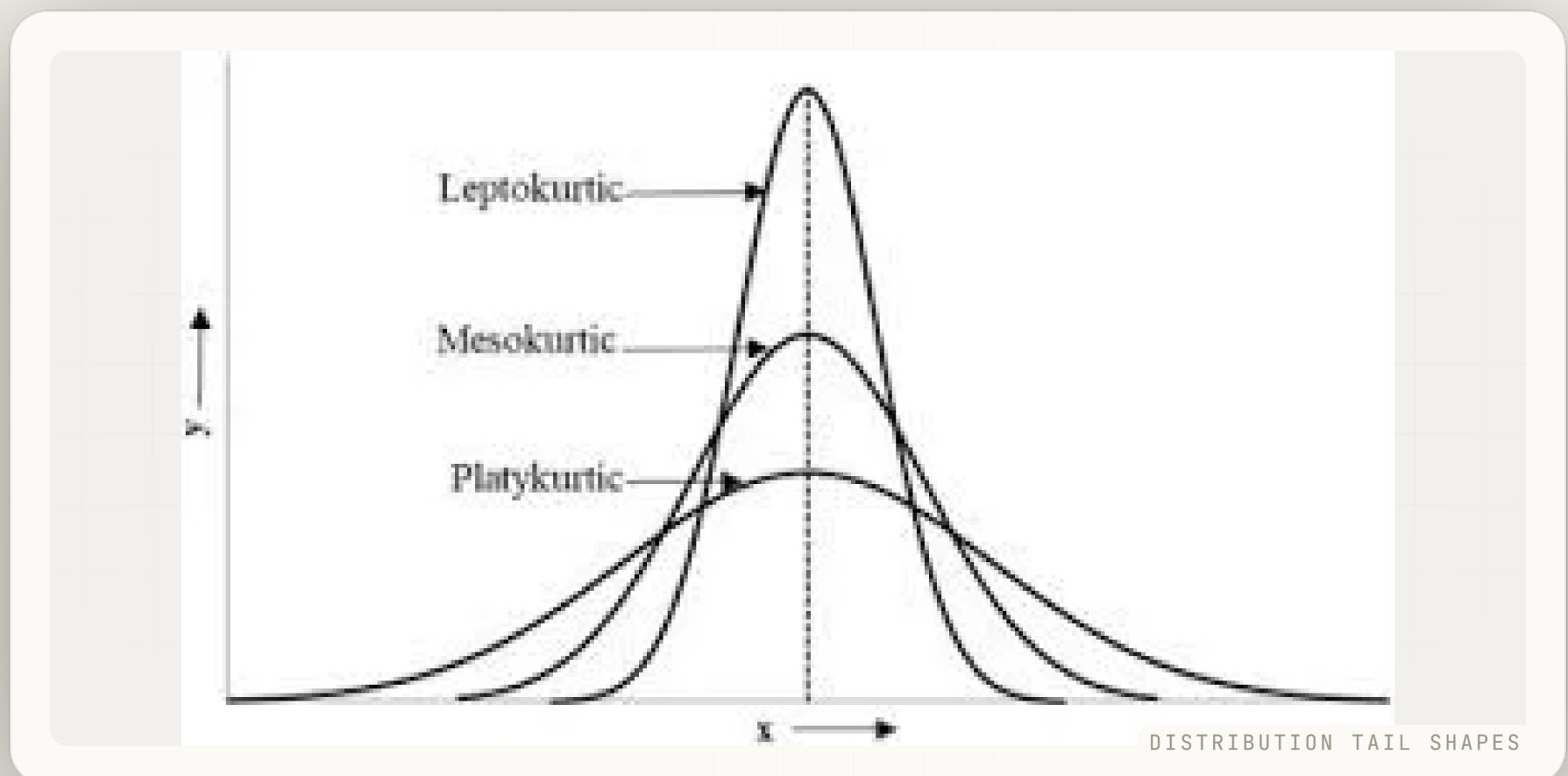
Kurtosis

tail weight • outlier risk

Kurtosis measures how much data clusters in the tails vs. the center. **High kurtosis means more extreme outliers** - critical to know before training on financial or sensor data.

USED IN

- Data analysis
- Feature engineering
- Anomaly detection



KNOWLEDGE GRAPH • KNOWLEDGE

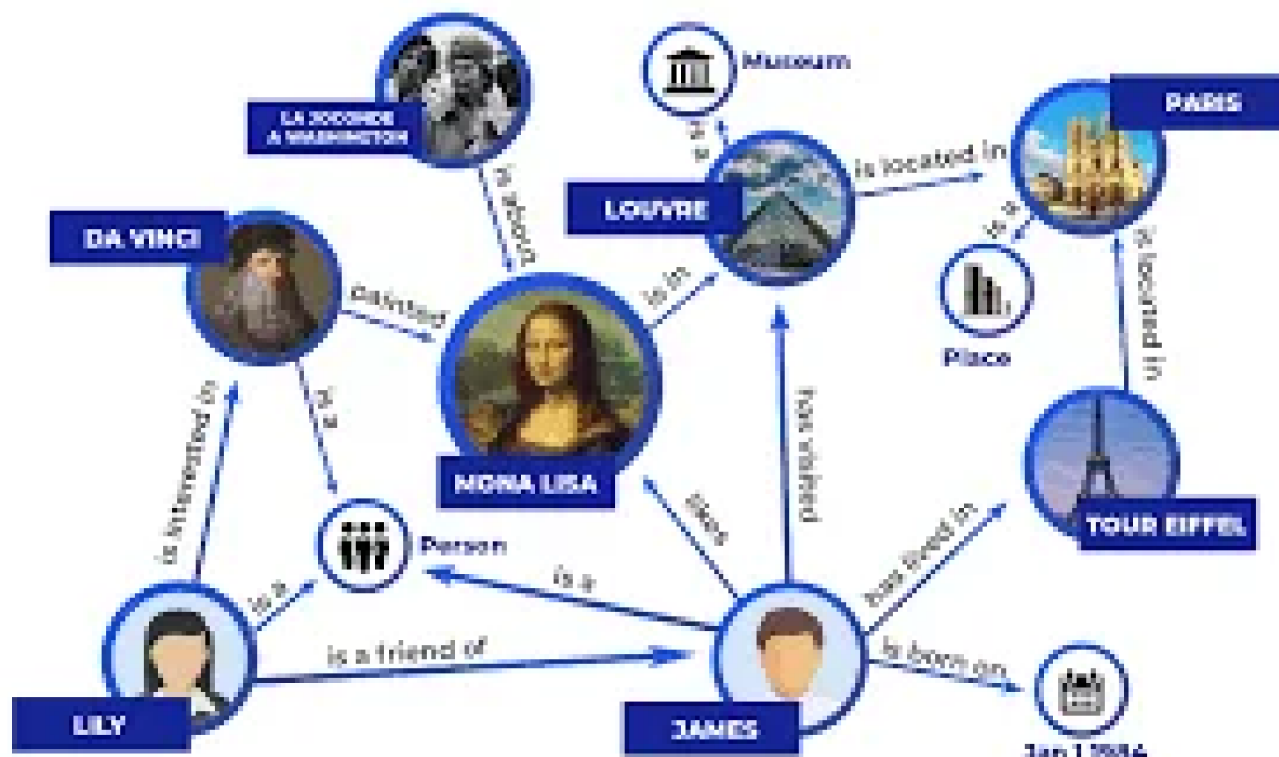
Knowledge Graph

entities • relationships • reasoning

A knowledge graph stores entities and their relationships as connected nodes and edges, letting AI systems **reason over structured knowledge** for search, QA, and recommendations.

USED IN

- RAG
- Search engines
- Recommender systems



K-NEAREST NEIGHBORS • LEARNING

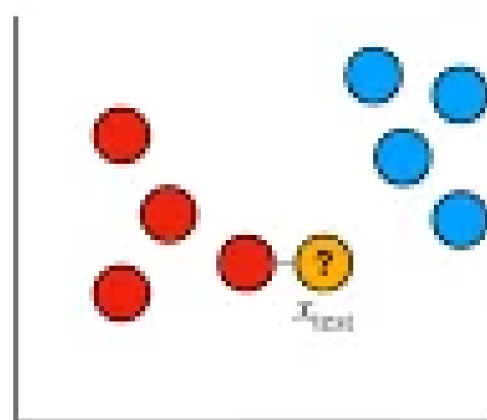
KNN

you are who your neighbors are

KNN classifies a point by looking at its K closest neighbors in the dataset - **no training required, just proximity**. Simple, powerful, and still widely used for recommendations and anomaly detection.

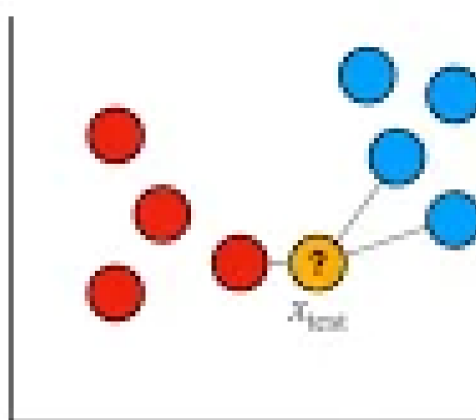
USED IN

- Classification
- Recommendations
- Anomaly detection



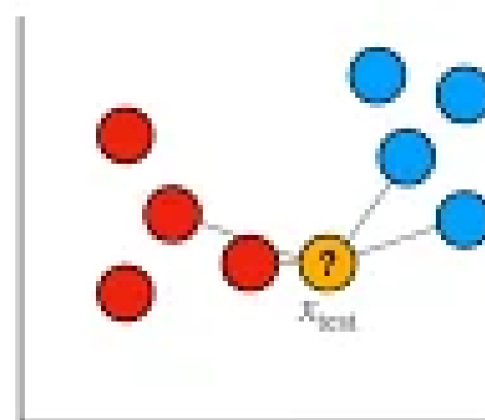
$k = 1$

Nearest point is **red**, so x_{test} classified as **red**



$k = 3$

Nearest points are (**red**, **blue**, **blue**) so x_{test} classified as **blue**



$k = 4$

Nearest points are (**red**, **red**, **blue**, **blue**) so classification of x_{test} is not properly defined

K=3 • NEAREST NEIGHBOR VOTE

KNOWLEDGE BASE • KNOWLEDGE

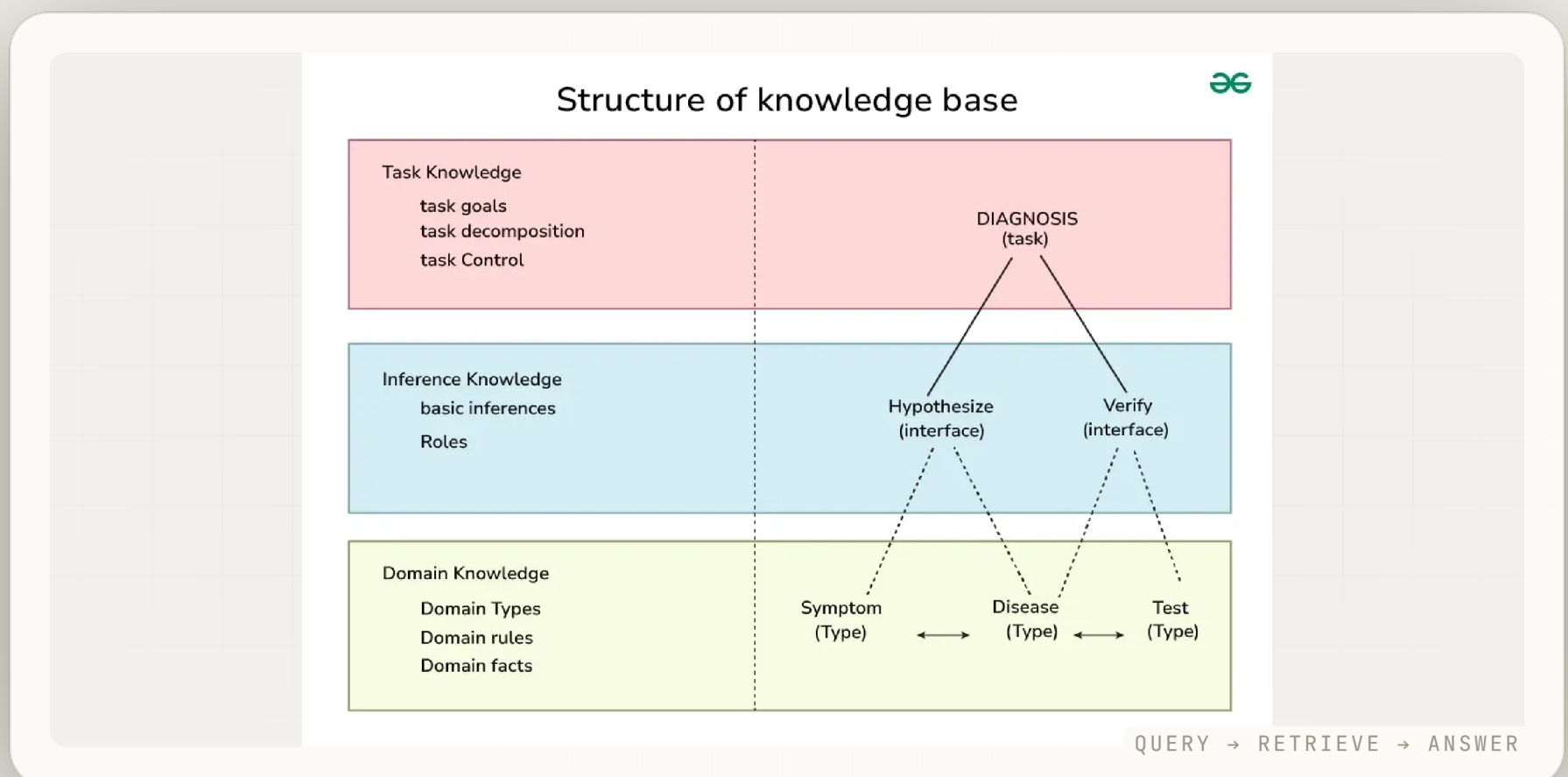
Knowledge Base

external memory • reliable facts

A knowledge base is a **centralized store of facts and documents** that AI systems pull from at inference time - the foundation of every RAG pipeline, chatbot, and intelligent assistant.

USED IN

- RAG pipelines
- Chatbots
- Enterprise search



KEY-VALUE CACHE • INFERENCE

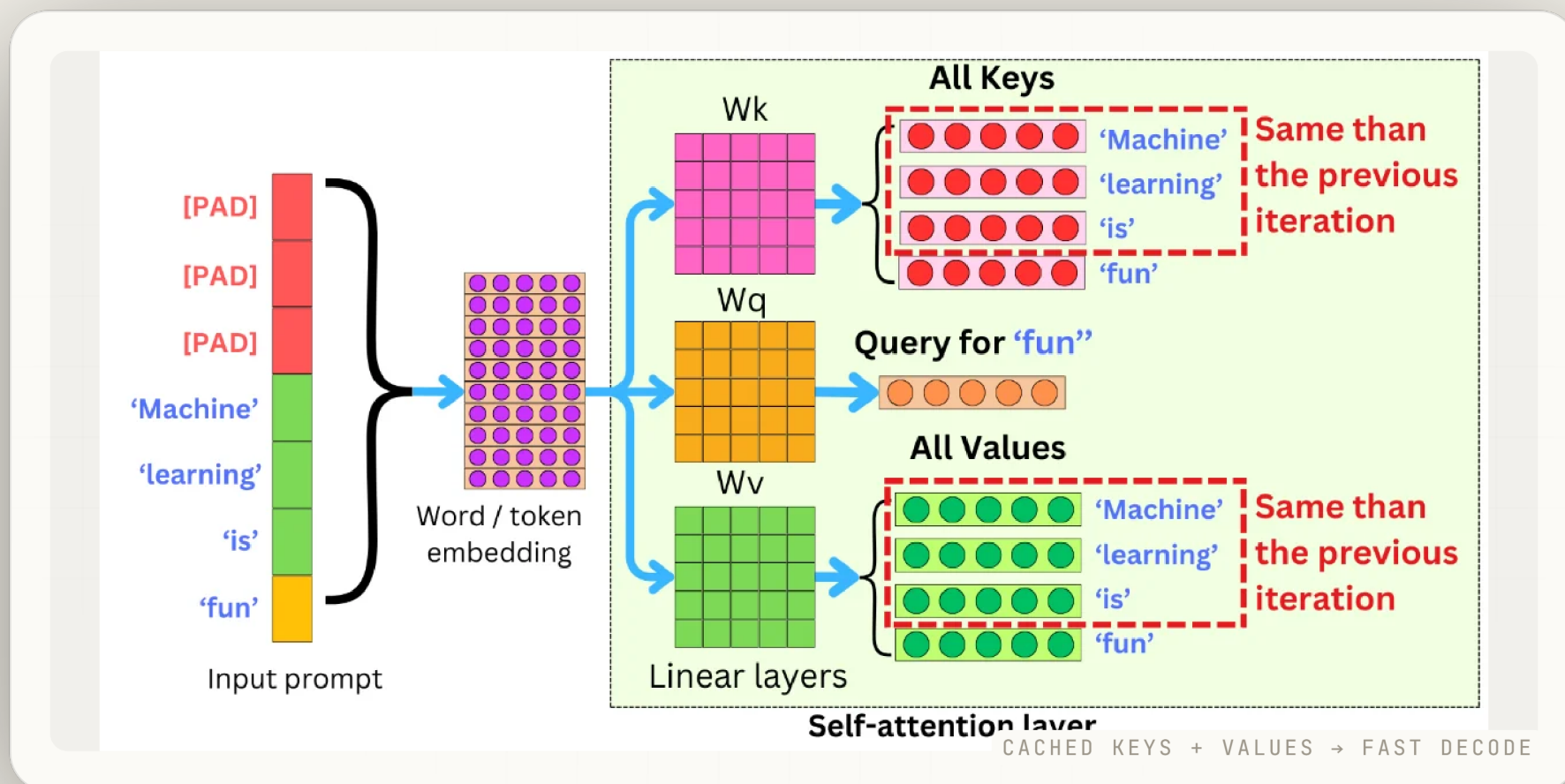
KV Cache

compute once • reuse forever

KV Cache stores the attention keys and values from earlier tokens so the model **never recomputes them again** - the single biggest trick behind fast LLM inference and low-latency chat.

USED IN

- LLM inference
- Fast generation
- Transformers



K-MEANS CLUSTERING • UNSUPERVISED

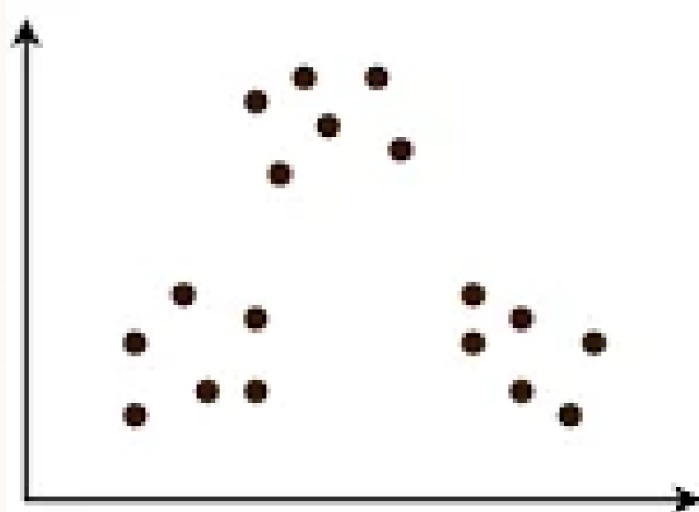
K-Means

find K groups • no labels needed

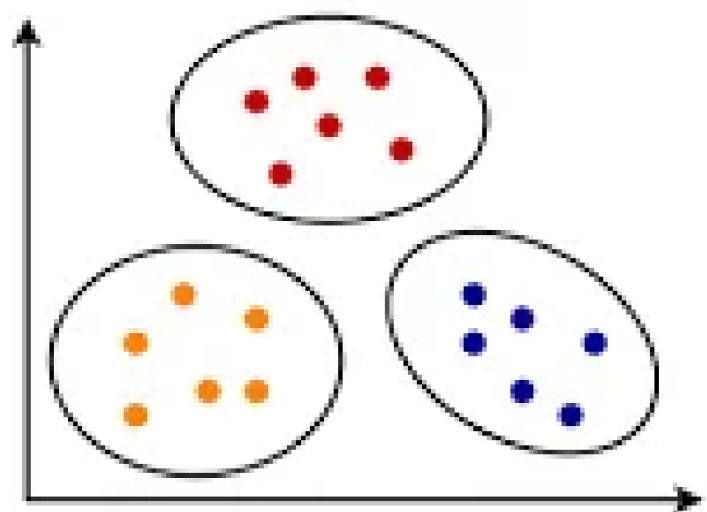
K-Means groups data into K clusters by iteratively **moving centroids until similar points converge** - the go-to algorithm for segmentation, topic modeling, and exploratory analysis.

USED IN

- Segmentation
- Topic modeling
- Exploratory analysis



Before K-Means



After K-Means

SCATTER → CENTROIDS → CLUSTERS

Kernels

similarity in disguise

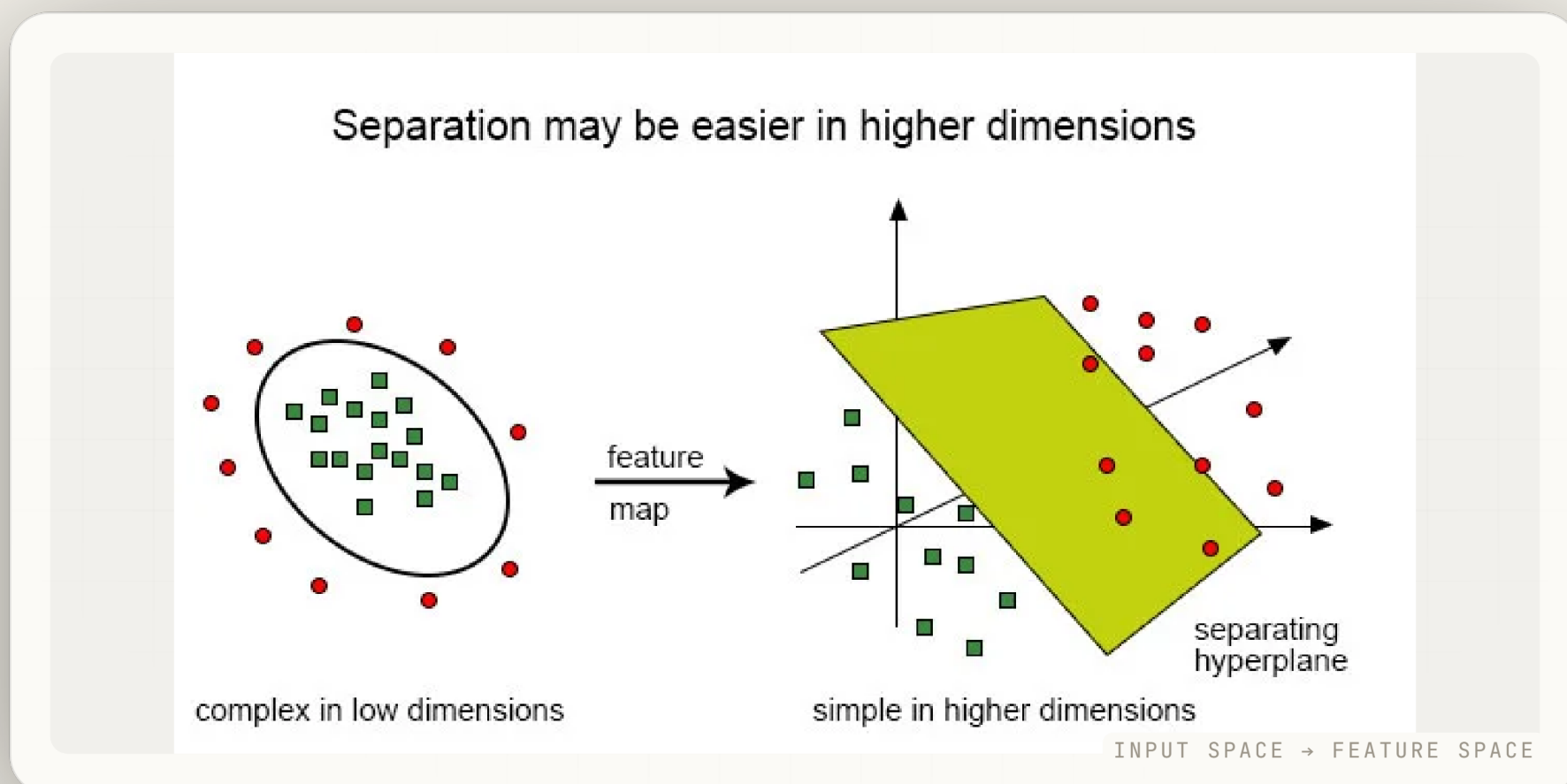
Kernels are functions that measure similarity between points in a **transformed high-dimensional space without ever computing it** - the math trick that makes SVMs handle non-linear data.

USED IN

• SVMs

• Gaussian processes

• Non-linear ML



KERAS • FRAMEWORK

Keras

deep learning made simple

Keras is the high-level API for TensorFlow that lets you **build, train, and deploy neural networks in a few lines** - the friendliest on-ramp to deep learning that exists.

USED IN

- Rapid prototyping
- Education
- TensorFlow workflows



APACHE KAFKA • INFRASTRUCTURE

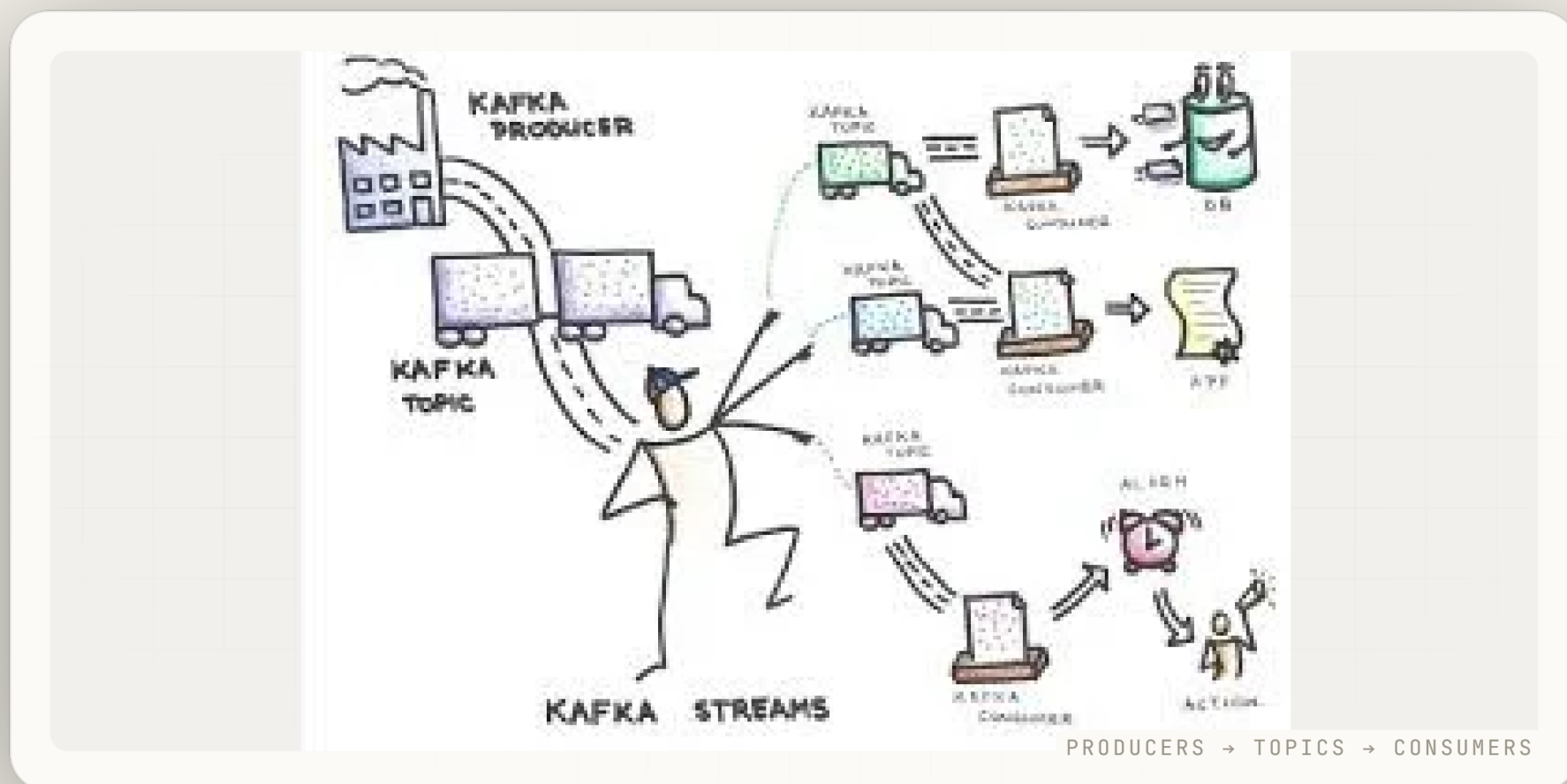
Kafka

real-time data • at any scale

Kafka is the distributed event-streaming backbone that **moves millions of events per second** to ML pipelines, feature stores, and monitoring systems - the plumbing of modern AI infrastructure.

USED IN

- ML data pipelines
- Real-time features
- Event monitoring



KUBERNETES (K8S) • MLOPS

Kubernetes

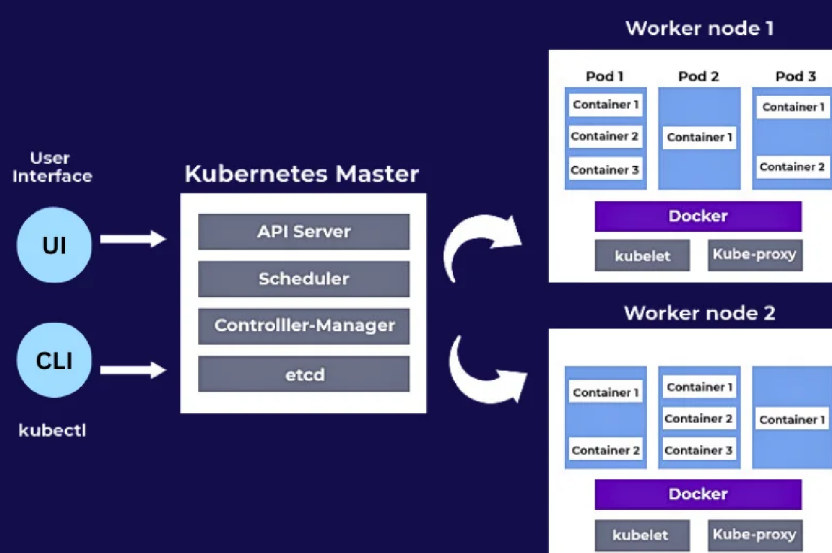
containers at scale • auto-everything

Kubernetes automates deploying, scaling, and managing containerized workloads - **the standard platform for serving ML models**, running training jobs, and managing AI infrastructure in the cloud.

USED IN

- Model serving
- MLOps pipelines
- Cloud AI infra

How Does Kubernetes Work?



RedSwitches

PODS • NODES • CLUSTER • SCALE

KNOWLEDGE DISTILLATION • COMPRESSION

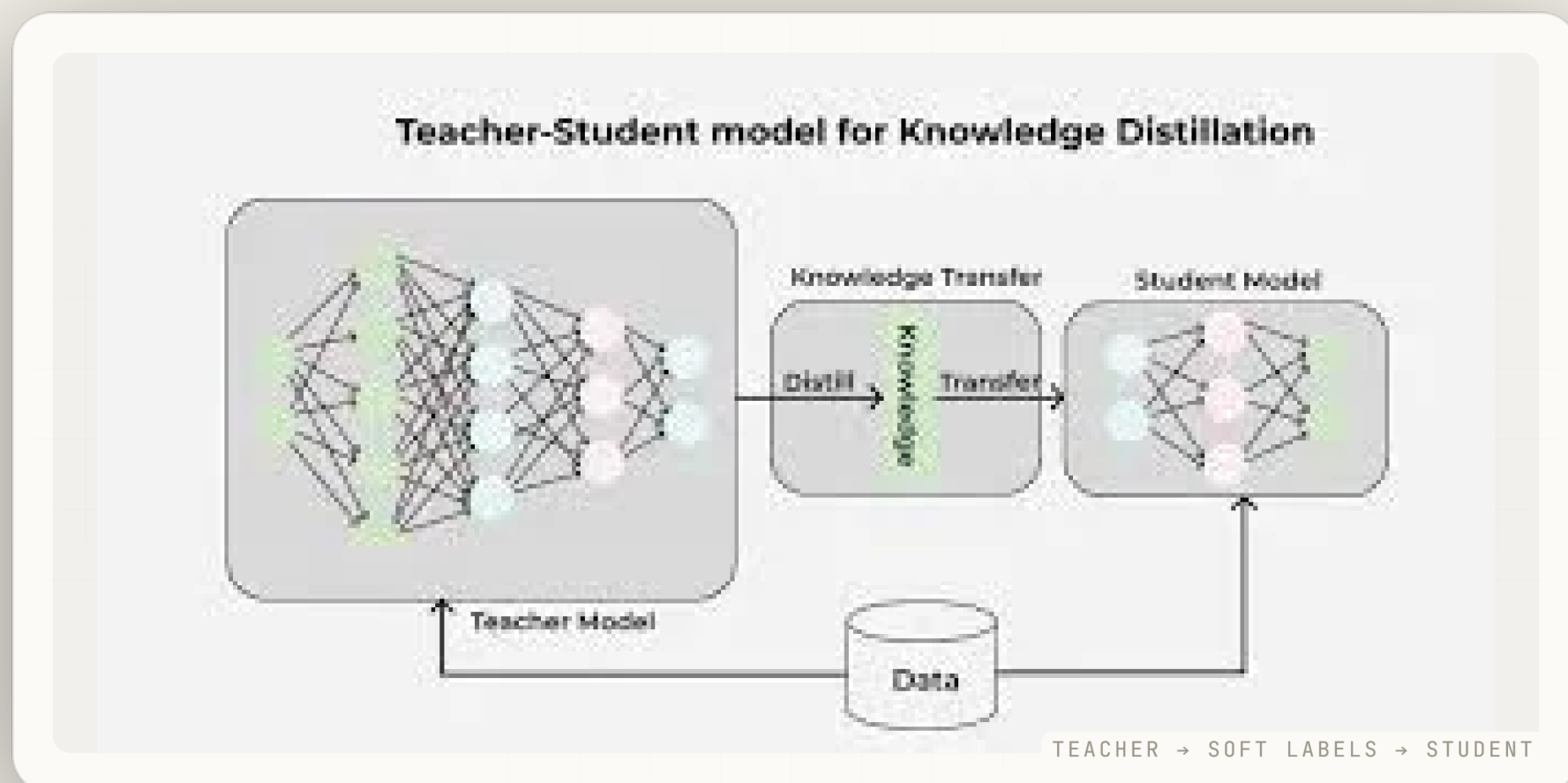
Knowledge Distillation

big teacher • small student

A large teacher model trains a smaller student model on its outputs - **transferring knowledge without transferring size**. The trick behind efficient models like DistilBERT and Phi.

USED IN

- Model compression
- Edge AI
- Efficient LLMs



↓ UP NEXT • COMPANIES & TOOLS

The K labs.

Four K-named platforms turning AI research into real products - from **frontier video generation** to the world's largest ML competition platform.



AI VIDEO GENERATION

Kling

text → video • by Kuaishou



Kuaishou's frontier **text-to-video AI** generating cinematic, temporally consistent video - one of the strongest challengers to Sora, Runway, and Veo in the video generation race.

Kuaishou

China's short-video giant

Text→Video

High-quality cinematic output

Frontier

Competing with Sora, Veo, Runway

WHAT IT BUILDS

High-quality video from text and images.

PARENT

Kuaishou Technology - owner of Kwai.

KNOWN FOR

Realistic motion and cinematic visuals.

FIND THEM

klingai.com

FRONTIER LLM FAMILY

Kimi



Moonshot AI • reasoning • long context

Moonshot AI's LLM series known for **strong reasoning, coding, and long-context understanding** - one of the most prominent frontier model families coming out of China.

Moonshot

Parent company, founded 2023

Long CTX

Extended context understanding

Reasoning

Coding and complex problem-solving

WHAT IT BUILDS

LLMs for coding, research, and agents.

PARENT

Moonshot AI, Beijing-based lab.

WHY IT MATTERS

China's frontier LLM competing globally.

FIND THEM

kimi.ai • @MoonshotAI

ML COMPETITION PLATFORM

Kaggle

datasets • competitions • community

kaggle



The world's largest ML platform - **datasets, competitions, notebooks, and a global community** of data scientists. Where careers are built, models are benchmarked, and skills are proven.

2010

Founded, acquired by Google
2017

15M+

Registered members
worldwide

Free

GPU notebooks, no credit card

WHAT IT OFFERS

Datasets, notebooks, competitions,
courses.

WHO USES IT

Students, researchers, and hiring
managers.

WHY IT MATTERS

The global benchmark for ML
practitioners.

FIND THEM

kaggle.com • @KaggleTeam

ML PLATFORM • MLOPS

Kubeflow



ML on Kubernetes • end-to-end

The open-source ML platform built on Kubernetes that manages **training, pipelines, tuning, and model serving** - the cornerstone of enterprise MLOps and cloud-native AI infrastructure.

Open

Open-source, CNCF project

K8s

Built on Kubernetes infrastructure

E2E

Train → tune → serve in one platform

WHAT IT DOES

End-to-end ML workflow management.

USED AT

Google, Spotify, Cisco, and large MLOps teams.

KEY FEATURES

Pipelines, hyperparameter tuning, serving.

FIND THEM

kubeflow.org • @kubeflow

↓ UP NEXT • PEOPLE TO FOLLOW

The K minds.

Five K-named educators, researchers, and builders -
worth a follow to keep your AI knowledge growing
and your career moving.

K

AI EDUCATOR

Krish Naik

YouTube • @krishnaik06



India's most-watched AI educator - **hands-on tutorials covering ML, deep learning, MLOps, and GenAI** that have helped millions build real-world AI skills from scratch.

1M+

YouTube subscribers

Hands-on

Project-based practical teaching

GenAI

Covers LLMs, agents, and MLOps

KNOWN FOR

End-to-end ML and GenAI project tutorials.

BEST FOR

Beginners building real AI skills fast.

WHY FOLLOW

Industry-focused, practical, no fluff.

FIND THEM

linkedin.com/in/naikkkrish •
@krishnaik06

NON-PROFIT EDUCATION PLATFORM

Khan Academy



free • world-class • Khanmigo

The nonprofit that made **world-class education free for everyone** - and now leads with Khanmigo, an AI tutor that personalizes learning for millions of students worldwide.

2008

Founded by Sal Khan

150M+

Registered learners worldwide

Khanmigo

AI-powered personalized tutor

WHAT IT OFFERS

Free courses in maths, science, CS, and AI.

REACH

Used in classrooms across 190+ countries.

THE AI PLAY

Khanmigo - an AI Socratic tutor for students.

FIND THEM

khanacademy.org • @khanacademy

DATA SCIENCE EDUCATOR

Ken Jee

YouTube • @KenJee1



One of the most-trusted voices for **data science career advice and ML education** – known for project walkthroughs, resume tips, and the inside track on breaking into the field.

Career

Top resource for DS career advice

Projects

End-to-end data science walkthroughs

Sports

Sports analytics background

KNOWN FOR

DS career content, projects, resumes.

BEST FOR

Breaking into data science and ML.

WHY FOLLOW

Real-world career and skills guidance.

FIND THEM

linkedin.com/in/kenjee • kennethjee.com

DEVOPS & CLOUD EDUCATOR

Kunal Kushwaha

YouTube • @kunal-kushwaha



Software engineer and open-source advocate who makes **cloud, DevOps, and system design accessible** - building the infrastructure skills that every modern AI engineer needs.

Open Source

Active contributor and advocate

Cloud

Kubernetes, DevOps, and system design

Community

One of the largest dev communities

KNOWN FOR

Cloud, DevOps, and open-source education.

BEST FOR

Developers learning Kubernetes and cloud.

WHY FOLLOW

Infrastructure skills for AI engineers.

FIND THEM

linkedin.com/in/kunal-kushwaha

AI RESEARCHER • RESNET

Kaiming He

MIT CSAIL • ResNet • MAE



Co-inventor of **ResNet**, the architecture that made very deep neural networks trainable - one of the most-cited AI papers in history. Now pushing the frontier at MIT CSAIL.

ResNet

Most cited CV paper in history

FAIR

Former Meta AI Research lead

MAE

Masked Autoencoders - CVPR 2022

KNOWN FOR

ResNet, skip connections, deep vision.

NOW AT

MIT CSAIL as assistant professor.

WHY FOLLOW

Foundational research on deep networks.

FIND THEM

people.csail.mit.edu/kaiming

The **K** stack, in one card.

Save it. Drop a comment with the K-term you use most.

01 **Kurtosis**

stats

02 **Knowledge Graph**

knowledge

03 **KNN**

learning

04 **Knowledge Base**

knowledge

05 **KV Cache**

inference

06 **K-Means**

clustering

07 **K-Fold CV**

evaluation

08 **Kernels**

math

09 **Keras**

framework

10 **Kafka**

infra

11 **Kubernetes**

MLOps

12 **Knowledge Distillation**

compression

+ **Kling • Kimi • Kaggle**

company

+ **Kubeflow**

MLOps

@ **Krish Naik • Ken Jee**

people

@ **Kunal • Kaiming • Khan**

people



FOLLOW FOR THE FULL A TO Z

Prasanna Kulkarni

in/prasanna-kulkarni-032650180

L Next